

NLP - IberLEF2026 HAHA: Humor Detection

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Abstract

Computational humor analysis remains a highly challenging yet rapidly evolving area within Computational Linguistics and Natural Language Processing. While significant progress has been made in automatic humor detection and classification through evaluation campaigns such as HAHA, the tasks of comprehensive humor recognition and generation are far from being fully resolved. This paper addresses the current challenges of processing figurative language by participating in the IberLEF 2026 HAHA evaluation campaign. We focus on problem of humor detection, specifically assessing the capabilities of Large Language Models (LLMs) to produce evaluating methods to automatically distinguish between human-authored and machine-generated humor. Our work aims to advance the understanding of automated humor mechanisms and contribute to the development of robust models for highly subjective NLP tasks.

1 Methodology

All code, model configurations, and replication scripts developed for our experiments are publicly available on GitHub.¹

2 Tasks

In this work, we focus on two core subtasks proposed in the IberLEF 2026 HAHA evaluation campaign, aimed at advancing the understanding of computational humor and machine-generated content:

- **Humor Detection:** determining if a news headline is a joke or not (satirical news headlines vs real news headlines). The main performance metric of this subtask will be the F1 score of the 'humorous' class. This subtask

is similar to the first subtask proposed in previous editions of the HAHA shared task, but this time it's applied to a particular domain where humorous and non-humorous content might sometimes be difficult to tell apart.

- **LLM-Generated Humor Detection:** determining if a joke inspired by a news headline was generated by an LLM or written by a human. The main performance metric for this subtask will be the F1 score of the 'automatic' class.

3 Dataset

The dataset provided for the evaluation campaign is distributed across multiple .tsv (Tab-Separated Values) files. All samples within the dataset are written in **Spanish**. Each individual sample represents a specific news headline accompanied by additional context.

The structure of the dataset consists of the following primary fields described in Table 1.

Field	Description
id	Unique identifier of the sample.
country	Country of origin where the headline was published.
date	Publication date of the headline.
headline	The main text of the news headline.
context	Additional text providing background information.

Table 1: Description of the main fields available in the dataset samples.

To monitor model performance and allow for rigorous evaluation, the data is split into exploratory (*trial*) and development (*dev*) sets for both tasks. The exact distribution of samples across the individual files is summarized in Table 2.

It is worth noting that the provided trial datasets (task1_trial.tsv and task2_trial.tsv) are exceptionally small, consisting of only 24 and 12 samples respectively. While these small subsets are highly valuable for initial pipeline validation, code

¹<https://github.com/Matejj2002/NLP-Project/tree/main>

File Name	Subtask	Number of Rows
task1_trial.tsv	Humor Detection	24
task1_dev.tsv	Humor Detection	540
task1_dev_gold	Humor Detection	
task2_dev_gold	Human AI	
task2_trial.tsv	Human AI	12
task2_dev.tsv	Human AI	350

Table 2: Data split sizes and sample distribution for Task 1 and Task 2.

debugging, and quick sanity checks of the models, their limited size poses a significant challenge for robust statistical generalization. Consequently, any performance metrics obtained on these trial sets must be interpreted with caution due to the high variance inherent to such small sample sizes. To mitigate this limitation, our primary local evaluation and hyperparameter tuning rely heavily on the larger development sets (task1_dev.tsv and task2_dev.tsv). After time, when competition got into post-evaluation phase, there were shared datasets for training with suffix gold, where was more labelled data.

4 Task 1: Humor Detection

4.1 Data Analysis

We performed a brief exploratory data analysis on the task1_trial.tsv dataset to identify basic structural properties of the text.

As shown in Figure 1(a), the trial dataset is perfectly balanced, containing exactly 12 real and 12 satirical headlines. Figure 1(b) illustrates that the dataset is evenly distributed among six Spanish-speaking nations (Colombia, Mexico, Puerto Rico, Spain, Uruguay, and Venezuela), with each country providing exactly 2 real and 2 satirical samples.

Furthermore, we analyzed the text length and linguistic features across classes:

- **Text Length:** Figure 2(a) shows that satirical headlines display a wider distribution in total character length compared to real news.
- **Word Length:** As visualized in Figure 2(b), the average word length for genuine news is strictly clustered around 5.2 characters, signaling formal vocabulary. Conversely, satirical text shows much higher variance (ranging from 3.5 to over 6.0 characters per word), reflecting a more flexible and creative use of language.

4.2 Approach 1: Stable Zero-Shot Ensemble

Given the limited size of the trial dataset, fine-tuning large-scale neural networks presents a severe risk of overfitting. To mitigate this issue, we developed a training-free **Zero-Shot Classification Ensemble** pipeline using multiple state-of-the-art cross-lingual and English models (including variants of mDeBERTa-v3, DeBERTa-v3, BART-large, and XLM-RoBERTa).

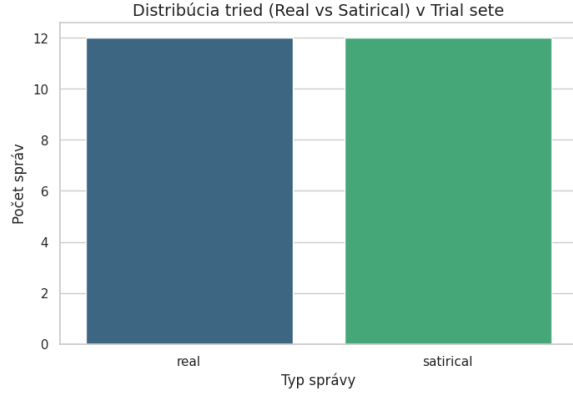
The core mechanics of this approach are designed to maximize stability and generalization across domains:

- **Prompt and Multiple Label Sets:** To capture the multi-faceted nature of humor, we evaluated each headline against three distinct label pairs mapping real vs. satirical text (e.g., “*serious journalism*” vs. “*ironic satire*”) using a standard hypothesis template: “*This text is {}.*”
- **Min-Max Score Normalization:** Since different models output raw confidence scores at varying scales, we applied *Min-Max Scaling* fitted on the trial distribution. This transformation guarantees that all models operate within a standardized range between 0 and 1.
- **Performance-Based Model Weighting:** Rather than using simple averaging, we evaluated each model independently on the trial set. Each model’s weight in the final ensemble was set proportional to its squared macro F1-score (F_1^2), heavily favoring top-performing architectures.
- **Threshold Optimization:** The final prediction is a weighted average of the normalized scores. The decision boundary was optimized using a grid search over the trial set, yielding an optimal threshold of **0.45**.

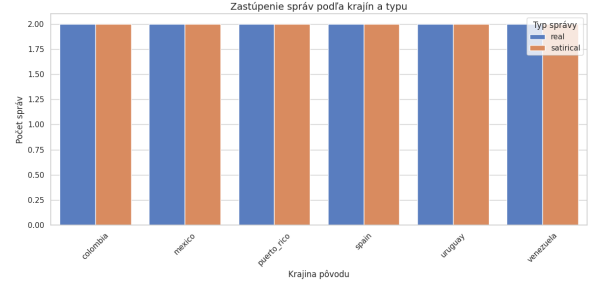
4.2.1 Experimental Results

The performance of the optimized Zero-Shot Ensemble was evaluated locally on both available datasets. On the initial pilot set (task1_trial.tsv), the ensemble achieved a well-balanced macro F1-score of **0.83**. The detailed classification report for this configuration is presented in Table 3.

When deployed on the competitive evaluation phase over the larger development set (task1_dev.tsv), the stable zero-shot ensemble

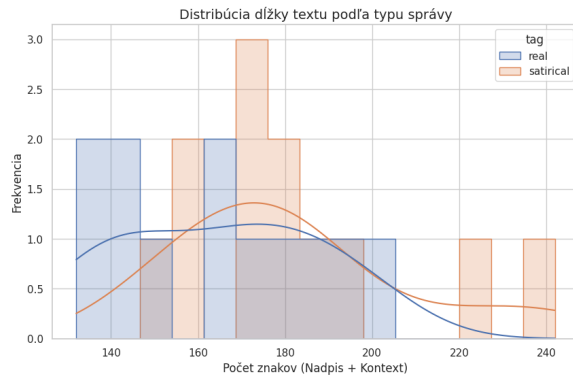


(a) Class Balance

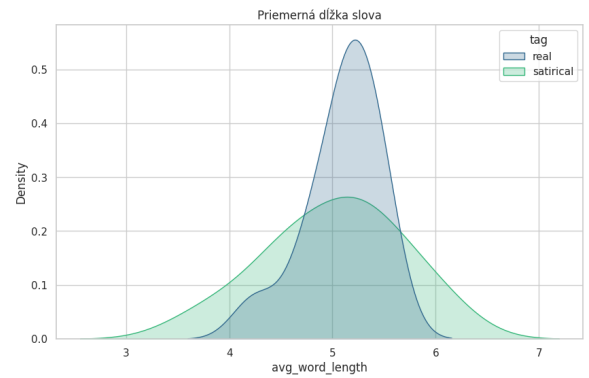


(b) Country Representation

Figure 1: Distribution of labels and geographic origins in the Task 1 trial set.



(a) Total Character Length



(b) Average Word Length

Figure 2: Analysis of text and word length properties between real and satirical classes.

Class	Precision	Recall	F1-Score	Support
real	0.83	0.83	0.83	12
satirical	0.83	0.83	0.83	12
Accuracy			0.83	24
Macro Avg	0.83	0.83	0.83	24
Weighted Avg	0.83	0.83	0.83	24

Table 3: Classification report on the trial dataset using the stable Zero-Shot Ensemble with an optimal threshold of 0.45.

successfully sustained its predictive power, achieving a final performance of **73%** accuracy on the leaderboard target. This proves that anchoring multiple zero-shot predictors with robust normalization acts as an excellent baseline for highly subjective NLP tasks without explicit training data.

4.3 Approach 2: Pseudo-Labeling and Transformer Fine-Tuning

To overcome the severe data scarcity of the trial set, we implemented a semi-supervised pipeline leveraging **Confidence-Based Pseudo-Labeling** combined with supervised transformer fine-tuning.

This approach allows us to exploit the larger, unlabelled development dataset to train a robust classification model.

The pipeline operates through the following key stages:

- **High-Confidence Selection:** We first utilize the soft probability scores generated by the stable Zero-Shot Ensemble (Approach 1) on the development set (`task1_dev.tsv`). To minimize the injection of noise, we apply a strict margin constraint of ± 0.15 around the optimal threshold ($t = 0.45$). Only samples falling outside this uncertainty zone (i.e., scores > 0.60 or < 0.30) are selected.
- **Dataset Augmentation:** The selected high-confidence samples are assigned their respective pseudo-labels (real or satirical) and merged with the original gold-standard trial dataset (`task1_trial.tsv`). This process scales up the available training data, making supervised training viable.

- **Model Architecture:** We adopt microsoft/mdeberta-v3-base as our core classifier. This model utilizes DeBERTa’s disentangled attention mechanism and is pre-trained via a multilingual masked language modeling objective, making it exceptionally proficient at capturing fine-grained stylistic nuances like satire and humor in Spanish.
- **Training and Validation Strategy:** The augmented dataset is tokenized with a maximum sequence length of 256 tokens. The network is fine-tuned for 3 epochs with a learning rate of 2×10^{-5} , weight decay of 0.01, and a batch size of 8. Crucially, to prevent the model from overfitting to potential errors in the pseudo-labels, we use the 100% clean trial dataset as our validation set, restoring the best performing checkpoint at the end of training.
- **Final Predictions and Overfitting:** The fine-tuned model was used to execute the final inference over the development set, generating the task1.tsv submission file. However, empirical evaluation revealed that despite a steady loss reduction during training, this approach suffered from severe **overfitting**. Due to the propagation of pseudo-label noise and the extreme scarcity of gold-standard data, the fine-tuned model failed to generalize well, ultimately delivering lower performance on the development set compared to the training-free Zero-Shot Ensemble.

4.4 Approach 3: Supervised Fine-Tuning on Post-Evaluation Gold Dataset

Following the initial competition phase, the release of the fully labeled development set (task1_dev_gold.tsv) provided a sufficient volume of high-quality training data. This enabled a purely supervised fine-tuning approach, completely bypassing the need for zero-shot heuristics or noisy pseudo-labeling.

To maximize the contextual information provided to the model, we concatenated the headline and context fields into a single unified text sequence. This combined dataset was then partitioned into an 85% training split and a 15% validation split to effectively monitor learning and prevent overfitting.

For this approach, we transitioned to the dccuchile/bert-base-spanish-wwm-cased ar-

chitecture (commonly known as BETO), a robust BERT model pre-trained exclusively on Spanish corpora. The dedicated Spanish pre-training makes this model highly adept at capturing the specific linguistic and cultural nuances required to detect satire in the target language.

The fine-tuning pipeline was executed with the following configuration:

- **Tokenization:** Sequences were padded and truncated to a maximum length of 128 tokens to optimize computational efficiency while retaining essential context.
- **Hyperparameters:** The network was fine-tuned for 4 epochs using a batch size of 16, a learning rate of 2×10^{-5} , and a weight decay of 0.01.
- **Optimization:** To ensure maximum generalizability, the training loop continually evaluated the validation loss at the end of each epoch, strictly restoring the weights of the best-performing model at the conclusion of the process.

4.4.1 Experimental Results

The fine-tuned model was used to generate final predictions on the task1_test.tsv dataset. By leveraging the high-quality gold data and a native Spanish language model, this supervised approach achieved an outstanding performance score of **82%**. This represents a substantial margin of improvement over both the Zero-Shot Ensemble and the Pseudo-Labeling techniques, definitively establishing it as the most successful and robust methodology among all our evaluated approaches.

5 Task 2: LLM-Generated Humor Detection

5.1 Data Analysis

The labeled data available for Task 2 is considerably more limited than for Task 1. The trial set contains only 12 samples, while the full gold-labeled development set (task2_dev_gold.tsv) provides 350 annotated examples. As both splits are required for training, we concatenate them to form a combined training set of **362 labeled examples**. The dataset is perfectly balanced, containing exactly 181 human and 181 machine samples. All jokes are written in Spanish and are paired with a corresponding news headline that served as the generation prompt.

Each sample in the dataset contains the fields described in Table 4.

Field	Description
id	Unique identifier of the sample.
headline	The news headline used as a prompt.
joke	The joke text, either human or machine-authored.
tag	Ground-truth label: human or machine.

Table 4: Description of the relevant fields in Task 2 dataset samples.

As shown in Figure 3(a), the training set is perfectly balanced with exactly 181 samples per class. Figure 3(b) reveals a notable structural difference: machine-generated jokes tend to be longer than human-written ones across all headline lengths, suggesting that LLMs produce more verbose outputs. This is further confirmed in Figure 3(c), where the total text length distribution of machine samples is shifted rightward and peaks around 220–250 characters, while human jokes are concentrated around 130–150 characters. Figure 3(d) shows that the average word length distributions largely overlap, indicating that lexical complexity alone is not a reliable distinguishing feature; length and verbosity are more informative signals.

5.2 Approach

5.2.1 Input Representation

A key design decision was to provide the model with both the joke and its originating headline simultaneously. Detecting machine-generated text is substantially easier when the model can compare the two: LLMs tend to produce jokes that follow the headline’s phrasing too closely or reproduce its structure in a stereotypical way. We therefore concatenate the two fields into a single input sequence

using a separator token:

headline [SEP] joke

This allows the encoder to attend jointly to both fields and model the relationship between them, which is not possible when the joke text is considered in isolation.

5.2.2 Loss Function

Although the class distribution is perfectly balanced, the competition metric is the F1 score of the machine class exclusively. To reflect this asymmetry, we apply a **weighted cross-entropy loss** that upweights the machine class by a factor of 1.3. This encourages the model to prioritize recall on the target class without severely degrading precision.

5.2.3 Training Configuration

All models were fine-tuned using the HuggingFace Trainer API with the following shared hyperparameter configuration:

- **Maximum sequence length:** 256 tokens
- **Batch size:** 16 (train), 32 (evaluation)
- **Learning rate:** 2×10^{-5} with a warmup ratio of 0.1
- **Weight decay:** 0.01
- **Maximum epochs:** 8, with early stopping (patience = 3)
- **Best checkpoint selection:** based on F1 of the machine class on task2_dev_gold.tsv

Local evaluation was performed on task2_dev_gold.tsv, which was *not* used for training, serving as a held-out development set to monitor generalization.

5.3 Experimental Results

We evaluated three progressively refined configurations, summarized in Table 5.

System	F1 (machine)
xlm-roberta-base (baseline)	0.8504
BETO fine-tuned	0.8671
Ensemble (BETO + xlm-roberta-large)	0.8678

Table 5: Task 2 results on the Codabench development set.

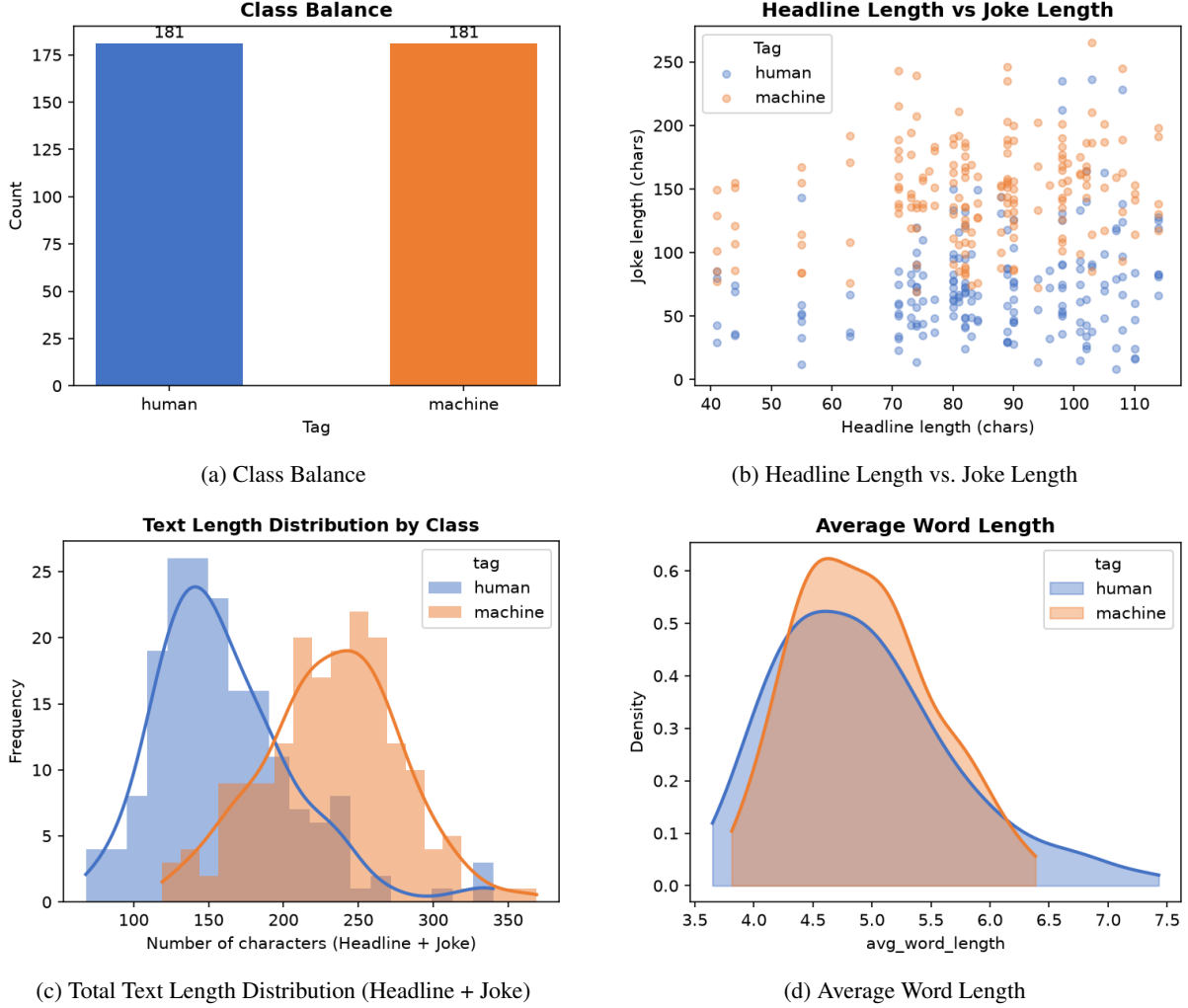


Figure 3: Distribution and text properties in the Task 2 training set.

Step 1 — Baseline: xlm-roberta-base. As a first experiment, we fine-tuned the multilingual xlm-roberta-base model on the combined 362-sample training set. Despite being trained on over 100 languages and having no specific adaptation to Spanish, this model achieved a strong baseline F1 of **0.8504** on the machine class, confirming that the headline–joke concatenation strategy provides a meaningful signal for the classifier.

Step 2 — Spanish-specific model: BETO. We then replaced the multilingual backbone with dccuchile/bert-base-spanish-wwm-cased (BETO), a BERT model pre-trained exclusively on large Spanish corpora using whole-word masking. The switch to a monolingual Spanish model delivered a consistent improvement of **+1.67 F1 points**, reaching **0.8671**. This confirms that language-specific pre-training is beneficial even for a binary detection task where surface-level

stylistic cues are the primary discriminating signal.

Step 3 — Cross-architecture ensemble: BETO + xlm-roberta-large. To further improve performance, we constructed a soft-vote ensemble combining the logits of BETO and xlm-roberta-large — two models differing substantially in pre-training data, tokenization strategy, and capacity. Despite this architectural diversity, the ensemble reached only **0.8678**, matching the marginal gain of +0.07 F1 points previously observed in the same-seed BETO ensemble. Both architectures converged to similar decision boundaries on this dataset, yielding highly correlated predictions and no meaningful complementarity.

5.4 Discussion

The dominant result in Task 2 is clear: the single most impactful intervention was replacing the multilingual backbone with a Spanish-specific model. The +1.67 F1 improvement from

xlm-roberta-base to BETO far outweighed any benefit from ensembling, regardless of whether the ensemble combined models trained with different random seeds or models of entirely different architectures.

The failure of ensembling to provide meaningful gains is best explained by the scale of the training data. With only 362 labeled examples, all fine-tuned models are likely converging to the same small set of surface-level cues that distinguish machine-generated from human-written jokes in this corpus — such as lexical predictability, structural regularity, or stylistic flatness. When models learn the same discriminating features, their errors are correlated by construction, and aggregating their predictions offers little benefit. A larger and more diverse labeled dataset would likely be a prerequisite for ensembling to produce measurable improvements on this task.

6 Conclusion

In this paper, we presented our system for the IberLEF 2026 HABA evaluation campaign, addressing two subtasks: humor detection in Spanish news headlines (Task 1) and detection of LLM-generated humor (Task 2).

For Task 1, we explored three approaches of increasing sophistication. A training-free zero-shot ensemble of cross-lingual models provided a solid baseline, achieving 0.73 F1 on the development set without any labeled training data. A semi-supervised pseudo-labeling approach failed to improve over this baseline due to noise propagation in the low-data regime. Finally, supervised fine-tuning of BETO on the released gold development data yielded the best result of **0.82 F1 on the humorous class** on the Codabench test set, demonstrating that a dedicated Spanish pre-trained model with sufficient labeled data substantially outperforms training-free alternatives.

For Task 2, the key finding was that language-specific pre-training matters more than model ensembling under data scarcity. Replacing the multilingual xlm-roberta-base with BETO yielded a +1.67 F1 improvement, while subsequent ensembling — both same-architecture multi-seed and cross-architecture — provided only marginal gains of +0.07 F1. Our best system achieved **0.87 F1 on the machine class** on the Codabench test set.

Across both tasks, the consistent lesson is that the quality of pre-trained representations in the tar-

get language is the dominant factor in low-resource settings, and that ensemble diversity cannot compensate for a training set that is too small for models to learn meaningfully different decision boundaries.